

PIERCE LUU

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Education

University of Toronto

Honours: Dean's List

Bachelor of Applied Science in Computer Engineering

Technical Skills

Languages: Python, C, C++, JavaScript, TypeScript, HTML/CSS, SQL (PostgreSQL)

Web: FastAPI, Flask, Node.js, Express.js, React, Next.js

AI, Data & ML: LangChain, LangGraph, NumPy, pandas, scikit-learn, PyTorch

Cloud & DevOps: Git, GitHub Actions, Docker, Kubernetes, AWS (EC2, S3, Lambda, DynamoDB, API Gateway, ECR, EKS, CloudWatch, Bedrock)

Experience

University of Toronto

May 2026 – Present

Undergraduate Researcher and Software Developer

Toronto, Ontario

- Architecting a web-based 3D virtual lab simulation platform for BME education at UofT using Node.js and Express, implementing a procedural step sequencer to deliver interactive safety training for BME students
- Established GitHub organization, CI/CD pipelines, and deployment infrastructure on Azure using GitHub Actions to support a multi-developer research software project

Ontario Online Schools

Jan 2026 – May 2026

Back-End Developer Intern

Remote

- Developed backend services using **Node.js** and **Express.js** for Ontario Online Schools' **Next.js** platform serving **40+ students**, integrating an AI chatbot using the **Anthropic API**, reducing support inquiries by **~65%**
- Configured **GitHub Actions CI/CD** pipelines to automate testing, building, and deployment workflows for a team of 4 engineers, reducing deployment time by **70%**
- Improved SEO across **9** pages by optimizing metadata and resolving Lighthouse issues, increasing SEO scores by **15%**

GamePill

Feb 2024 – June 2024

Game Development and Quality Assurance Intern

Concord, Ontario

- Refined gameplay systems debugging **10+ issues** and contributing to rapid prototypes in **Unreal Engine (C++)**.
- Documented **150+ bugs** and improvement suggestions, helping accelerate development iteration by **15%**

Projects

Traffer

- Engineered a traffic detection pipeline serving a **PyTorch** object detection model via **FastAPI** on **Kubernetes** (Minikube), achieving sub-**200ms** inference and **60%** Docker image reduction through multi-stage builds
- Automated CI/CD running **40 Pytest** unit tests, cutting deployment time to **under 3** minutes
- Packaged MLOps utilities for metrics reporting, batch inference, and model versioning, achieving **95%+** code coverage

TeddyTalk - 2x Winner at Canada's Largest Hardware Hackathon (MakeUofT)

- Developed a **Python** backend using **OpenCV** for face detection and fine-tuned a FER+ based emotion recognition model on custom data, achieving **95% accuracy** for real-time inference
- Won 1st Place Best Use of Arduino UNO Q and Best Use of ElevenLabs API out of **69 teams**

Silo Labs - Cursor for Embedded Systems

- Built an agentic firmware-generation platform powered by a **10-node** LangGraph multi-agent pipeline that turns requests into pin assignments, clock configurations, compiled C drivers, and a live hardware simulation
- Parallelized per-peripheral feature branches via **LangGraph** Send fan-out with a custom state reducer, and instrumented every node to stream real-time token, cost, and latency metrics over SSE to the frontend
- Grounded agent reasoning in a structured hardware knowledge graph built from chip specifications, eliminating hallucinated low-level configurations by resolving every decision through validated tool calls